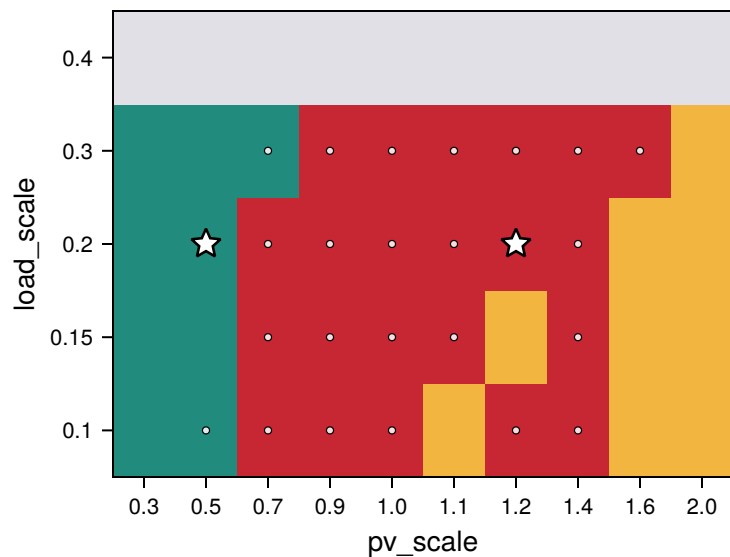


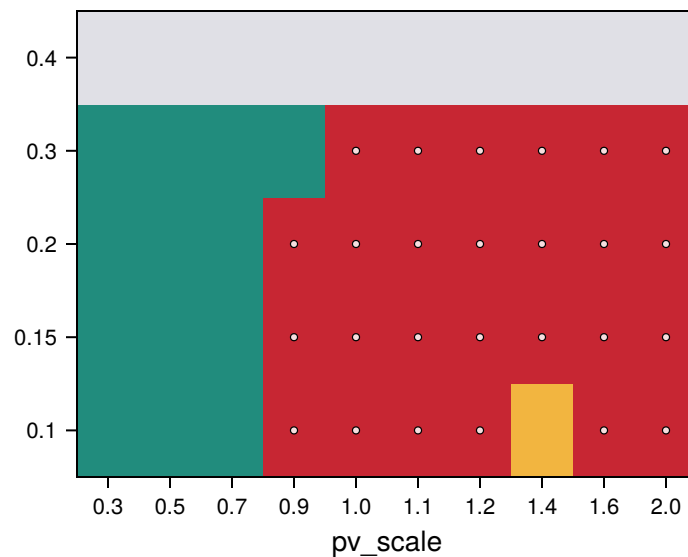
Where the SOC branch-flow relaxation is EXACT

3-bus high-PV fixture ($r=x=0.05$, $v_{\min}=0.95$) · T=24 · priced export ON · free cone-gap detector, no AC oracle

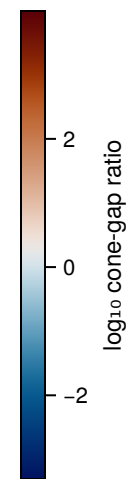
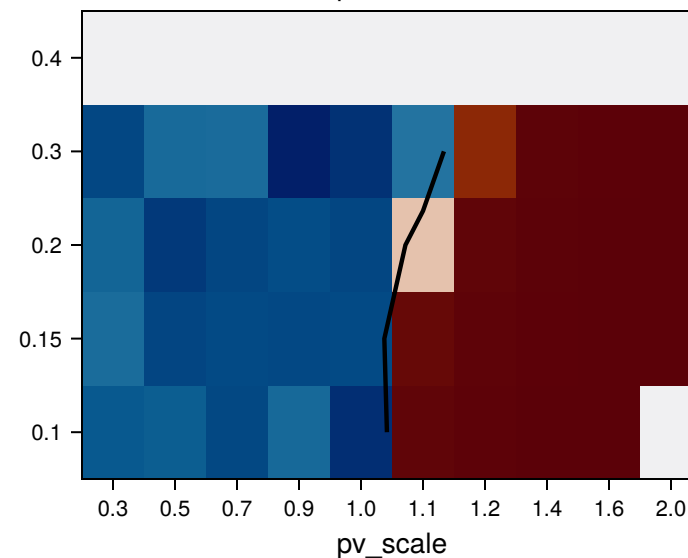
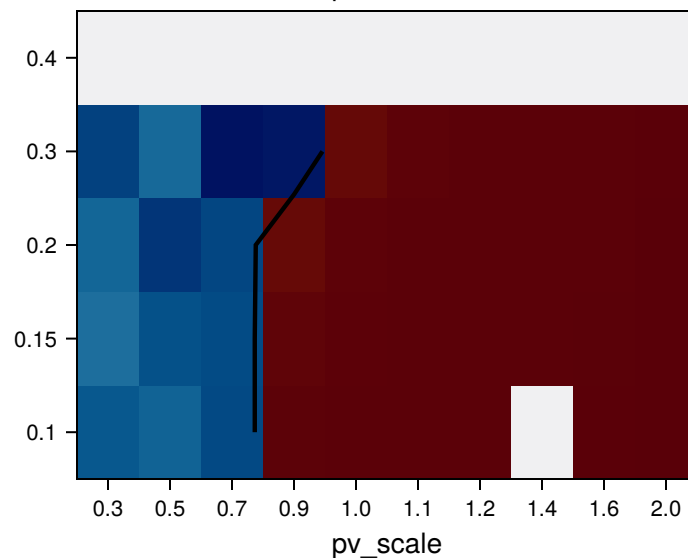
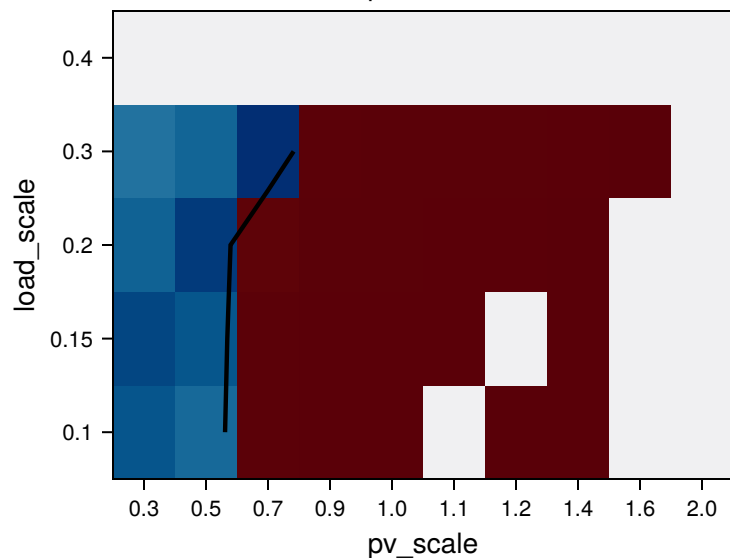
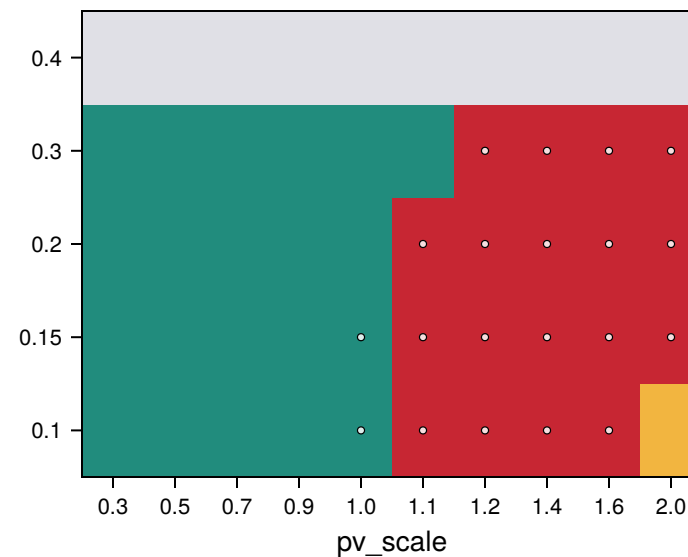
vmax = 1.05



vmax = 1.075



vmax = 1.1



exact (ratio ≤ 1) INEXACT (ratio > 1) guard tripped — UNMEASURED infeasible — unserveable $\circ$ voltage bound active $\star$ EXACT-04 control point

Row 2 black line = classification threshold (ratio = 1). Solved points are bimodal: max exact ratio 0.024, min inexact ratio 8.45 — a $\sim 350\times$ empty band, so the boundary's position does not depend on where the threshold is placed.